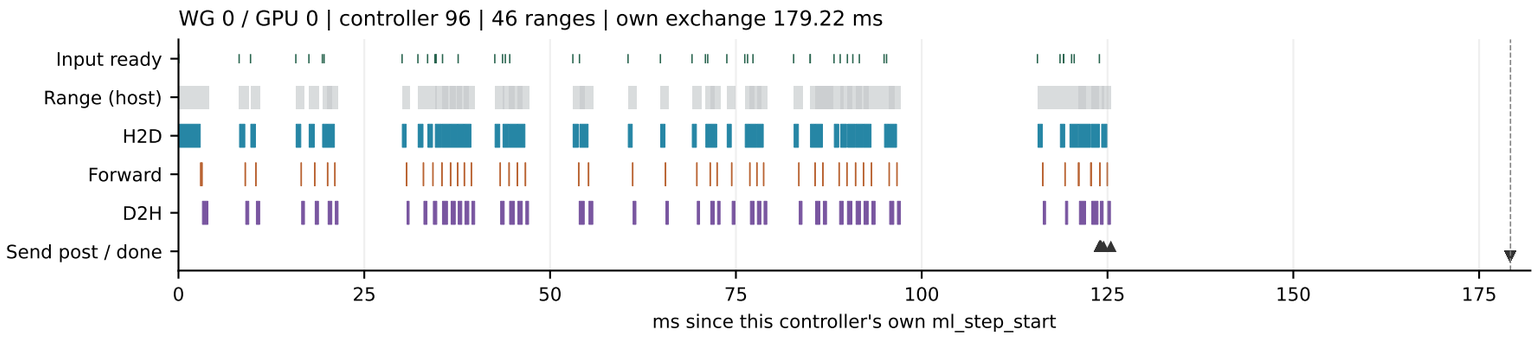
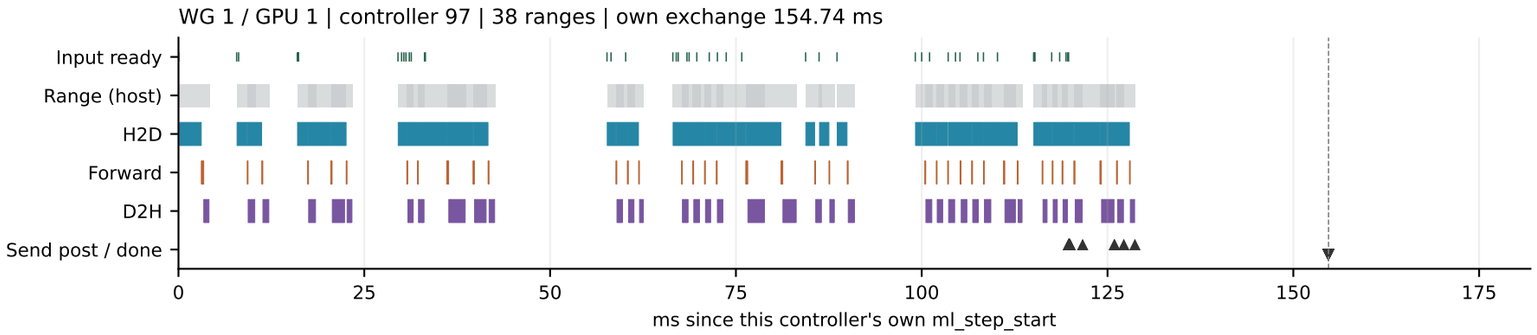
mlp | 96p96c4g | default | collective vs p2p1 | rep 1 | call 6 / MAIA step 189

Overall MAIA ML-step makespan: collective 349.01 ms | p2p1 221.49 ms



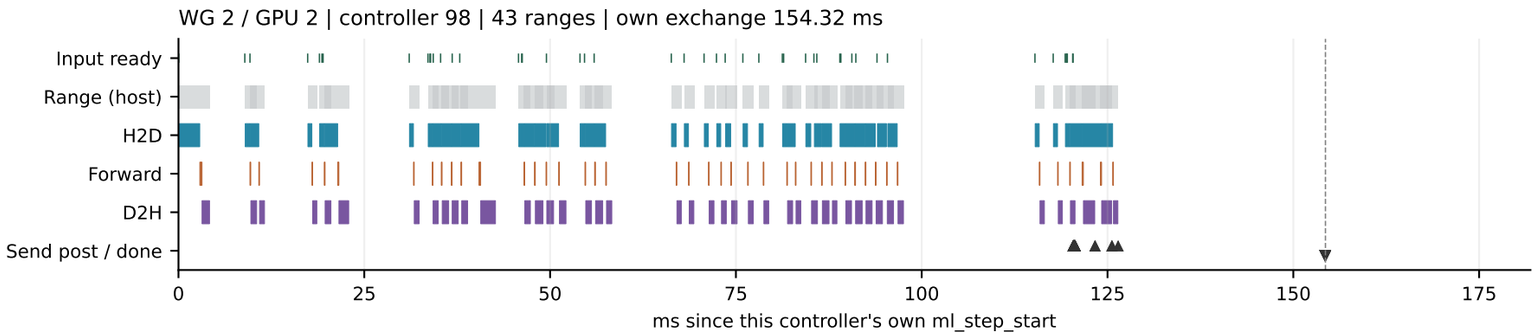
GPU work (ms)	Collective	p2p1
H2D	25.940	38.757
Forward	2.605	4.803
D2H	12.685	22.234
GPU SUM	41.231	65.794

Controllers: coll 96 / P2P 96
Coll host device elapsed: 55.321 ms
P2P GPU envelope: 125.276 ms
Different scopes; not an elapsed-time comparison.



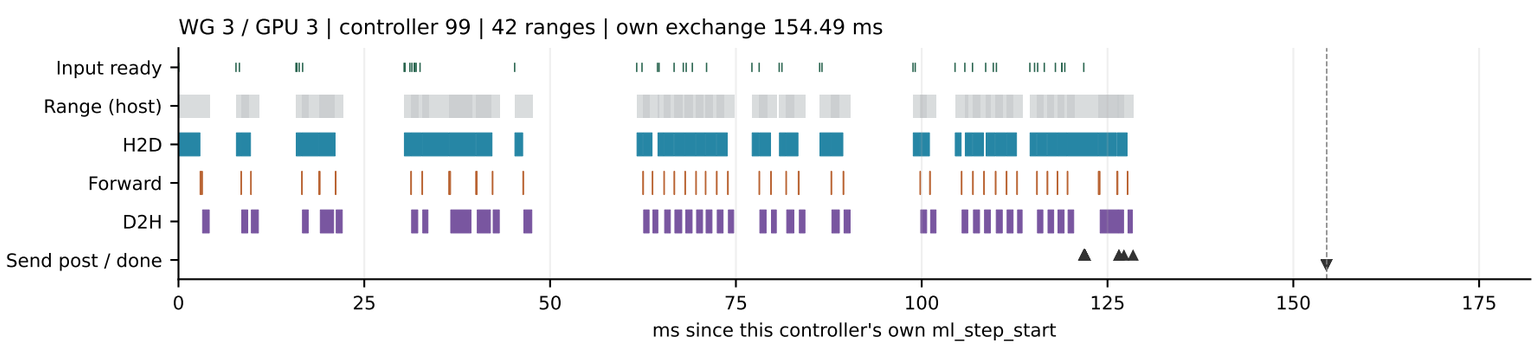
GPU work (ms)	Collective	p2p1
H2D	49.462	72.309
Forward	2.602	5.376
D2H	19.346	35.096
GPU SUM	71.410	112.780

Controllers: coll 97 / P2P 97
Coll host device elapsed: 71.579 ms
P2P GPU envelope: 128.496 ms
Different scopes; not an elapsed-time comparison.



GPU work (ms)	Collective	p2p1
H2D	22.773	46.187
Forward	2.605	4.799
D2H	24.532	35.163
GPU SUM	49.911	86.148

Controllers: coll 98 / P2P 98
Coll host device elapsed: 50.055 ms
P2P GPU envelope: 126.284 ms
Different scopes; not an elapsed-time comparison.



GPU work (ms)	Collective	p2p1
H2D	24.425	63.182
Forward	2.601	5.271
D2H	24.322	39.146
GPU SUM	51.347	107.598

Controllers: coll 99 / P2P 99
Coll host device elapsed: 51.550 ms
P2P GPU envelope: 128.238 ms
Different scopes; not an elapsed-time comparison.

Each controller has its own zero; rows are NOT globally synchronized. CUDA-event durations are reliable; host-anchored inter-request overlap is approximate. GPU work SUM is not elapsed time or a critical path.